

Homework 10

40 points

1. Box A has 2 red, 3 blue. Box B has 4 red, 1 blue. You choose a box at random (equal chance) and draw one ball. It is red. What is the probability you picked Box B? **3 points**

• A : 2/5 red

B : 4/5 red

• Bayes:
$$P(B|R) = \frac{P(R|B) P(B)}{P(R)}$$

$$= \frac{4/5 \cdot 1/2}{2/5 \cdot 1/2 + 4/5 \cdot 1/2}$$

$$= \frac{2}{3}$$

2. A single die is rolled. What is the probability of getting an even number, given that a number greater than 3 is rolled? **3 points**

- $P(\# > 3) = 1/2 \rightarrow 4, 5, 6$

- $P(\text{Even} \mid \# > 3) = 2/3 \rightarrow 4, 6$

3. A DNA test has the following accuracy:

- True match rate 99%
- False match rate 1%

Only **1 in 500 people** could be the true match.

Someone matches. What is probability they are the true match? **4 points**

$$\begin{aligned} \bullet P(TM | M) &= \frac{0.99(1/500)}{0.99(1/500) + 0.01(499/500)} \\ &= \frac{0.00198}{0.01196} = 0.1655 \end{aligned}$$

4. You have two coins: A fair coin and a double-headed coin. You pick a coin randomly and get Heads. Find the probability that the coin is double-headed.

3 points

$$\bullet P(DH|H) = \frac{P(H|DH) P(DH)}{P(H)}$$

$$= \frac{1 \left(\frac{1}{2} \right)}{1 \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{1}{2} \right)} = \frac{2}{3}$$

5. An insurance company knows that a customer has a 2% chance of filing a \$10,000 claim this year. What is the expected payout per customer?
If the company wants an expected profit of \$200 per customer, what should be the premium? **3 + 2 = 5 points**

- $E(\text{Payout per C}) = 0.02(10000) = \200

- $\text{Premium} - 200 = \text{Profit}(200)$

$$\text{Premium} = 200 + 200 = \$400$$

6. A slot machine costs \$2 per play. Payouts:

Prize	Amount	Probability
Jackpot	\$500	1/2000
Small win	\$20	1/50
No win	\$0	remainder

Find the expected net gain per play. **5 points**

• Avg payout

$$E[X] = 500 \left(\frac{1}{2000}\right) + 20 \left(\frac{1}{50}\right) = 0.65$$

• Avg net gain

$$E[\text{net gain}] = 0.65 - \text{cost}(2) = -1.35 \text{ per play}$$

7. A factory wants the sample average weight of packages to be within ± 0.2 kg ^{margin} of the true mean with probability 0.95. Package weights have standard deviation $\sigma = 1.5$ kg. Using CLT, how large should the sample n be? **6**

points

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\bullet z @ 0.95 = 1.96$$

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$\bullet P(|\bar{X} - \mu| \leq 0.2) = 0.95$$

$$P(|z| \leq \frac{0.2}{\sigma/\sqrt{n}}) = 0.95$$

$$P(|z| \leq 1.96) = 0.95$$

$$\bullet \frac{0.2}{\sigma/\sqrt{n}} = 1.96$$

$$\sqrt{n} = \frac{1.96 \cdot 1.5}{0.2}$$

$$n = \left(\frac{1.96 \cdot 1.5}{0.2} \right)^2 = 216.1 \rightarrow 217$$

8. Let $X_1, \dots, X_{64}$ be independent and identically distributed with mean $\mu = 10$ and variance $\sigma^2 = 16$. Approximate the probability that the sample mean $\bar{X}$ exceeds 11. **5 points**

$$\sigma = \sqrt{16} = 4, \mu = 10, n = 64$$

$$\bullet \sigma_{\bar{x}} \text{ (sample mean)} = \frac{4}{\sqrt{64}} = 0.5$$

$$\bullet z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{11 - 10}{0.5} = 2$$

$$\bullet P(\bar{X} > 11) = P(Z > 2) = 1 - P(Z \leq 2) \\ = 1 - 0.9773 = 0.0228$$

9. Ten lightbulbs lasted (in hours):

2800, 2900, 3050, 3150, 2700, 3000, 2990, 2850, 3100, 2950. — x

Find a **95% CI** for the mean lifetime (assume normality). **6 points**

- $Sum = 29490, \quad n = 10$

$$\bar{x} = 2949$$

- $\sum (x - \bar{x})^2 \rightarrow 174090$

- $df = n - 1 = 9$

$$S = \sqrt{\frac{174090}{9}} = 139.05$$

- $CI = \bar{x} \pm t \frac{S}{\sqrt{n}} = 2949 \pm 2.262 \frac{139.05}{\sqrt{10}}$
 $t_{9, 0.975}$

$$= (2949 - 99.5, 2949 + 99.5)$$

$$= (2849.5, 3048.5)$$